

Cool, Chill, and Cruel: Probabilistic Ranking with Greedy Approach for Recovering Sparse Binary Secrets in LWE

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Abstract—Sparse binary LWE secrets are widely used in homomorphic encryption (HE) and post-quantum cryptography due to their efficiency. However, recent attacks (e.g., SALSA, PICANTE, VERDE) show that ML and statistical methods can break such secrets at moderate dimensions. Classical lattice-based attacks reveal that after partial reduction, some LWE matrix columns remain high-variance (“cruel”), while others become low-variance (“cool”).

We introduce a three-zone approach—Cruel, Chill, and Cool—that refines this model. By combining brute force for cruel columns, ML-based probabilistic enumeration for chill columns, and greedy bit-flips for cool columns, we achieve a more flexible attack strategy. We analyze complexity, present schematic diagrams, and explore potential implications for RLWE, demonstrating that a three-region partition better exploits partially reduced matrices compared to prior two-region methods.

I. INTRODUCTION

A. Problem Statement and Existing Attacks

Cryptanalysis of sparse LWE secrets typically involves:

- **Primal and Dual attacks:** Dimensionality reduction and short-vector recovery [4].
- **Hybrid meet-in-the-middle (MiTM):** Partial bit guessing combined with lattice reduction [1].
- **ML-based attacks** (e.g., SALSA, PICANTE, VERDE): Identifying correct bits via neural networks and transformers [2], [3], [7].

These methods classify columns as either “hard” (high variance) or “easy” (low variance). However, in practice, variance decreases gradually, forming a **sloping** region that doesn’t fit into a strict two-zone model.

B. Our Contribution

We propose a refined three-zone approach—**Cruel, Chill, and Cool**—for sparse LWE key recovery:

- **Cruel region:** High-variance columns ($\sim \frac{q}{12}$), requiring brute force or meet-in-the-middle.
- **Chill region:** Moderate-variance columns handled via probabilistic or ML-based enumeration, reducing brute-force overhead.
- **Cool region:** Low-variance columns, recoverable via greedy bit-flips.

This partitioning minimizes the search space in each zone. We provide complexity estimates, diagrams, and

discuss RLWE implications, showing how this three-region method improves over traditional two-region attacks by leveraging ML-assisted partial recovery.

II. RELATED WORK

A. Existing Lattice-Based Attacks

Several known attacks on small or sparse secrets include:

- Primal, Dual, and Hybrid attacks [1], [4].
- SALSA, a statistical attack leveraging machine learning [2].
- PICANTE, which incorporates transformer-based heuristics for LWE key recovery [7].
- SALSA VERDE, an improved version of SALSA, refining its statistical recovery approach [3].

B. Inspiration for Our Work

This work was inspired by the recent research presented in [5], which explored the effects of partial lattice reduction on LWE secrets and motivated the three-zone partitioning approach.

C. Conclusion

We introduce a novel three-zone attack strategy that extends existing binary LWE key recovery methods by partitioning reduced columns into ****Cruel, Chill, and Cool**** regions. This allows for an adaptive attack leveraging brute-force, probabilistic heuristics, and efficient greedy methods. Our analysis suggests that this approach can improve over traditional two-region models by better handling variance gradients in partially reduced lattices. Future work includes exploring its applicability to structured LWE variants such as Ring-LWE.

III. PRELIMINARIES AND BACKGROUND

A. Learning With Errors (LWE)

LWE is defined by sampling:

- $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ (uniform),
- $\mathbf{s} \in D_s \subseteq \mathbb{Z}_q^n$ (secret),
- $\mathbf{e} \in D_e \subseteq \mathbb{Z}_q^m$ (small noise).

The challenge is to recover $\mathbf{s}$ from:

$$\mathbf{b} \equiv \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q}.$$

For ****sparse binary LWE****, $\mathbf{s} \in \{0, 1\}^n$ has Hamming weight h .

a) *Variants.*: Modifying D_s, D_e leads to binary, ternary, Gaussian, or ring-based (RLWE) variants.

B. Partial Lattice Reduction

LWE is often attacked via partial lattice reduction using the q -ary embedding:

$$\Lambda = \begin{pmatrix} 0 & qI_n \\ \omega I_m & \mathbf{A} \end{pmatrix},$$

where ω balances reduction strength vs. noise amplification. Applying BKZ, flatter, or polish yields:

$$\mathbf{A}' = \mathbf{R}\mathbf{A}, \quad \mathbf{b}' = \mathbf{R}\mathbf{b}.$$

Reduction quality is measured by:

$$\rho = \frac{\sigma(\mathbf{A}')}{\sigma(\mathbf{A}_{\text{init}})}, \quad \sigma(\mathbf{A}_{\text{init}}) = \frac{q}{\sqrt{12}}.$$

Lower ρ indicates better reduction but may increase noise.

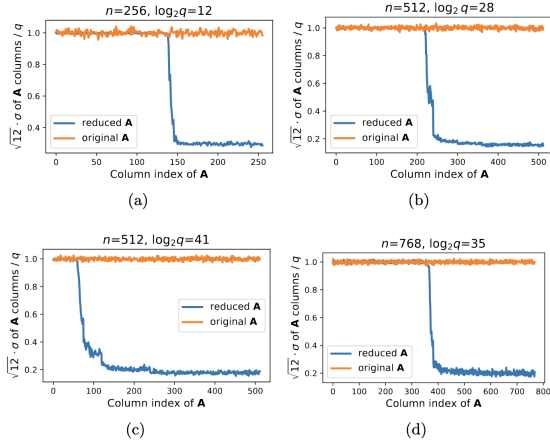


Fig. 1: Variance separation: original $\mathbf{A}$ (orange) vs. reduced $\mathbf{A}'$ (blue), with a drop at index n_u .

a) *Sub-Sampling & Interleaving.*: Often, m is sub-sampled from $4n+$ LWE samples. Reduction techniques like BKZ and flatter may be interleaved, with termination based on ρ convergence.

C. Greedy Recovery of “Cool” Bits

Following [5], after solving the high-variance (“cruel”) subset, the remaining low-variance (“cool”) columns can be recovered efficiently using greedy bit-flips. We summarize this as **Algorithm 1**.

Interpretation. We individually check whether flipping a particular bit from 0 to 1 lowers the residual’s variance; if so, we keep that flip. This linear-time method works well when each column has significantly reduced variance, ensuring a strong statistical signal per bit.

a) *Limitations.*: When a subset of columns has only moderately reduced variance (“sloping”), flipping bits individually can fail if noise remains too large. Hence, our proposal in Section V extends beyond a pure cruel/cool partition, introducing a “chill” region for partial or probabilistic enumeration.

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Algorithm 1 Greedy Recovery of Cool Bits (from [5])

Require: $\mathbf{s}^*$ partially known (correct cruel bits); the rest initially = 0.

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1: for  $i \in \{n - n_r, \dots, n - 1\}$  do
2:    $s^*[i] \leftarrow 0$ 
3:    $\sigma_0 \leftarrow \sigma(\mathbf{A} \cdot \mathbf{s}^* - \mathbf{b} \bmod q)$ 
4:    $s^*[i] \leftarrow 1$ 
5:    $\sigma_1 \leftarrow \sigma(\mathbf{A} \cdot \mathbf{s}^* - \mathbf{b} \bmod q)$ 
6:   if  $\sigma_0 \leq \sigma_1$  then
7:      $s^*[i] \leftarrow 0$ 
8:   else
9:      $s^*[i] \leftarrow 1$ 
10:  end if
11: end for

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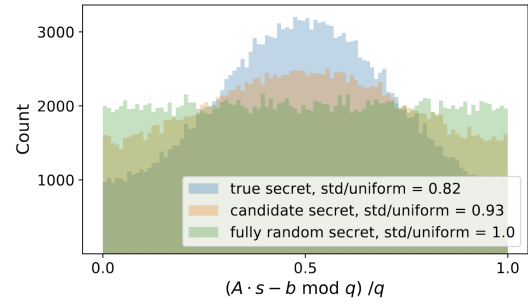


Fig. 2: Residual histograms mod q . **Blue:** True secret guess (narrow distribution), **Orange:** Partially correct guess (e.g., cruel bits only), **Green:** Random guess (near-uniform).

IV. VARIANCE DIAGRAM AND MOTIVATION FOR A THREE-ZONE PARTITION

A. Empirical Observation

After applying partial lattice reduction to the LWE matrix $\mathbf{A}$, the column-by-column standard deviation often exhibits a near-vertical “cliff”. However, we observe that in some cases, the transition between high variance and low variance can form a sloping region, shown in Figure 3.

a) *Why a Slope Can Appear.*: Depending on how flatter, BKZ, or other reduction algorithms interleave, the final column norms might not transition abruptly from $\approx q/\sqrt{12}$ to a fully reduced state. Instead, a fraction of columns may exhibit intermediate “semi-reduced” variance, producing what we refer to as a sloping region.

B. Why a Three-Zone Partition?

Historically, “Cool & Cruel” style attacks assumed columns are either:

- **Cruel:** High variance $\approx q/\sqrt{12}$, needing brute force or meet-in-the-middle;
- **Cool:** Low variance, quickly recovered via bit-flip or minimal overhead.

However, a purely two-zone approach misses the subtle middle region where variance is neither clearly high

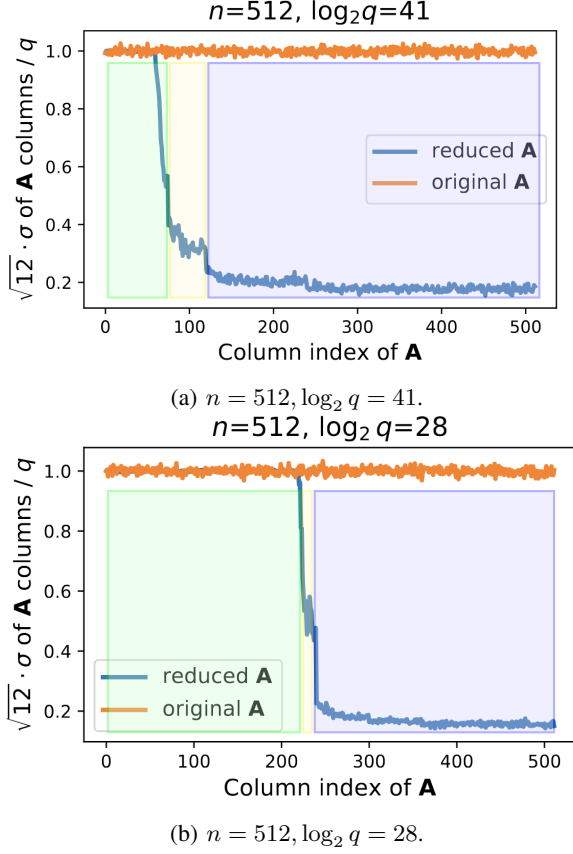


Fig. 3: Empirical plots of column standard deviation (scaled by $\sqrt{12}/q$) before and after reduction, annotated with **Cruel** (green), **Chill** (yellow), and **Cool** (blue) regions. In some instances (left subfigure), the transition from high variance to low variance is relatively narrow, while in others (right subfigure), it forms a longer slope.

nor fully reduced. We propose a three-zone decomposition:

$$\begin{aligned} \mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}} &\subseteq \{1, \dots, n\}, \\ \mathcal{I}_{\text{cruel}} \cup \mathcal{I}_{\text{chill}} \cup \mathcal{I}_{\text{cool}} &= \{1, \dots, n\}. \end{aligned}$$

- **Cruel (green):** High-variance columns ($\mathbf{A}_{*,j} \sim q/\sqrt{12}$), requiring brute force or hybrid guessing.
- **Chill (yellow):** Transitional columns with moderate variance, handled via probabilistic or ML-based methods.
- **Cool (blue):** Low-variance columns, recoverable via greedy bit flips.

Next section formalizes the details of each zone's attack strategy.

V. PROPOSED METHOD: COOL, CHILL, AND CRUEL

A. Threshold-Based Zoning

After partially reducing the LWE matrix $\mathbf{A}'$ (Section III-B), each column j has an associated standard deviation σ_j . We define two thresholds T_c and T_h with

$$T_c > T_h > 0,$$

for instance $T_c \approx 0.8\sigma_u$ and $T_h \approx 0.3\sigma_u$, where $\sigma_u \approx \frac{q}{\sqrt{12}}$ is the original uniform standard deviation. Then, each column j is classified into one of three regions:

$$\text{Region}(j) = \begin{cases} \text{Cruel}, & \text{if } \sigma_j \geq T_c, \\ \text{Chill}, & \text{if } T_h \leq \sigma_j < T_c, \\ \text{Cool}, & \text{if } \sigma_j < T_h. \end{cases}$$

This yields three disjoint subsets $\mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}} \subseteq \{1, \dots, n\}$.

B. Attack Phases

Our attack proceeds in three phases based on column variance:

- **Phase 1: Brute Force (Cruel)** – High-variance columns $\mathcal{I}_{\text{cruel}}$ are guessed via brute-force or meet-in-the-middle:

$$\mathbf{s}_{\text{cruel}} = \text{BruteForceCruel}(\mathbf{A}_{*,\mathcal{I}_{\text{cruel}}}, \mathbf{b})$$

with $\text{cruel_ones} = \text{HW}(\mathbf{s}_{\text{cruel}})$ denoting the recovered 1-bits.

- **Phase 2: Probabilistic Enumeration (Chill)** – Medium-variance columns $\mathcal{I}_{\text{chill}}$ are ranked by probability:

$$p_j = \text{EstimateProb}(j, \mathbf{A}, \mathbf{s}_{\text{cruel}})$$

and $k = h - \text{cruel_ones}$ bits are assigned via ranking, with local flips used if refinement is needed.

- **Phase 3: Greedy Bit-Flip (Cool)** – Low-variance columns $\mathcal{I}_{\text{cool}}$ are solved with GREEDYFLIP, flipping bits to minimize residual deviation.

C. Pseudocode: Multi-Zone Recovery

Algorithm 2 Multi-Zone Attack: Cool, Chill, and Cruel

Require: Reduced matrix $\mathbf{A}$, vector $\mathbf{b}$, modulus q , Hamming weight h

- 1: $(\mathcal{I}_{\text{cruel}}, \mathcal{I}_{\text{chill}}, \mathcal{I}_{\text{cool}}) \leftarrow \text{ClassifyColumns}(\mathbf{A}, T_c, T_h)$
- 2: $\mathbf{s}_{\text{cruel}} \leftarrow \text{BruteForceCruel}(\mathbf{A}_{*,\mathcal{I}_{\text{cruel}}}, \mathbf{b})$
- 3: $\text{cruel_ones} \leftarrow \text{HW}(\mathbf{s}_{\text{cruel}})$
- 4: $p_j \leftarrow \text{EstimateProb}(j, \mathbf{A}, \mathbf{s}_{\text{cruel}}) \forall j \in \mathcal{I}_{\text{chill}}$
- 5: $k = h - \text{cruel_ones}, \mathbf{s}_{\text{chill}} \leftarrow \text{RankAndAssign}(p_j, k)$
- 6: $\mathbf{s}_{\text{cool}} \leftarrow \text{GreedyFlip}(\mathbf{A}, \mathbf{b}, \mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}})$
- 7: $\text{RefineChill}(\mathbf{A}, \mathbf{b}, \mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}}, \mathbf{s}_{\text{cool}})$
- 8: **return** $(\mathbf{s}_{\text{cruel}}, \mathbf{s}_{\text{chill}}, \mathbf{s}_{\text{cool}})$

a) Implementation Notes.:

- **ESTIMATEPROB** ranks chill columns using variance or ML-based probability estimation.
- **RANKANDASSIGN** selects the top k chill bits as 1, refining with flips if needed.
- **GREEDYFLIP** minimizes residual deviation by adjusting low-variance bits.

Each region is handled according to its variance, balancing brute force, ML ranking, and greedy recovery to optimize efficiency and success probability.

VI. MATHEMATICAL AND LOGICAL DETAILS

A. Probability and Complexity Insights

Given a classification into Cruel, Chill, and Cool regions with respective counts $n_{\text{cruel}}, n_{\text{chill}}, n_{\text{cool}}$, we define:

$$\text{cruel_1} = \text{HW}(\mathbf{s}_{\text{cruel}}), \quad k = h - \text{cruel_1}.$$

where h is the total Hamming weight. The remaining k bits must be assigned within $\mathcal{I}_{\text{chill}}$ and $\mathcal{I}_{\text{cool}}$.

To determine which $\mathcal{I}_{\text{chill}}$ columns contain 1s, each column $j \in \mathcal{I}_{\text{chill}}$ is ranked by:

$$p_j = \Pr[s_j = 1],$$

where p_j can be estimated heuristically ($p_j \propto 1/\sigma_j$), via a small ML model (logistic regression or neural net on residuals), or through direct statistical tests. The top $k_{\text{chill}} \leq k$ ranked columns are set to 1, with the remainder placed in the cool set.

If the initial classification is incorrect, a local search (BFS or iterative bit-flip) refines borderline assignments without fully enumerating $\binom{n_{\text{chill}}}{k_{\text{chill}}}$.

a) *Complexity Estimation.*: The attack cost combines:

$$2^{n_{\text{cruel}}} + \binom{n_{\text{chill}}}{k_{\text{chill}}} \times \text{ML_overhead} + O(n_{\text{cool}}),$$

where the brute force term dominates for large n_{cruel} , while the chill region overhead depends on partial enumeration and ranking efficiency.

Notional Complexity Formula.

$$\mathbf{C}(n, k) \approx 2^{n_{\text{cruel}}} + \binom{n_{\text{chill}}}{k_{\text{chill}}} \times \text{ML_overhead}(1) \times \left[1 + \text{liability_coef} \times (n - n_{\text{cruel}} - n_{\text{chill}}) \right].$$

b) *Term-by-Term Explanation:*

- 1) $2^{n_{\text{cruel}}}$: Exhaustive or meet-in-the-middle enumeration for high-variance columns.
- 2) $\binom{n_{\text{chill}}}{k_{\text{chill}}}$: Enumerations for selecting 1-bits in the chill region, mitigated by ranking or BFS.
- 3) $\text{ML_overhead}(1)$: Overhead per candidate guess, from ML or statistical checks.
- 4) $\text{liability_coef} \times (\dots)$: Models local flips or re-runs in the chill region.
- 5) **Cool overhead**: Typically $O(n_{\text{cool}})$, from bit-flip verification in a reduced domain.

c) *Comparison to Original Cool-and-Cruel:* In [5], columns are labeled as **hard** (n_u brute-force bits) or **easy** (linear-time flips), with complexity:

$$2^{n_u} + O(n - n_u),$$

where $n_u \approx n_{\text{cruel}}$.

This method struggles with *medium-variance columns*, while our **three-zone** approach lowers n_{cruel} via partial enumeration ($\binom{n_{\text{chill}}}{k_{\text{chill}}}$) or ML, reducing brute-force cost at the expense of chill-region overhead.

VII. DISCUSSION, FUTURE WORK, AND CONCLUSION

A. Discussion

Our three-zone (Cruel, Chill, Cool) approach refines partial lattice reduction by isolating a Chill region to handle medium-variance columns. We aim to reduce search space while maintaining efficiency. Section VI showed how Chill enumerations replace excessive brute force at some overhead cost.

a) *Balancing Lattice Reduction.*: Stronger reduction lowers n_{cruel} but amplifies noise σ_e , affecting Chill/Cool classification [5]. An adaptive approach—tuning penalty parameter ω and BKZ/flatten block sizes—may balance enumeration complexity and error impact.

B. Future Work

a) *1. Ternary/Gaussian Secrets.*: Extending Chill classification to ternary or Gaussian secrets requires adjusting probability estimates or separate ML models. Partial reduction depth may complicate recovery for multi-valued secrets.

b) *2. Integrating SALSA or Transformers.*: ML-based methods (e.g., SALSA, PICANTE, VERDE) could improve Chill classification. However, managing ML overhead is critical to prevent excessive complexity increases.

c) *3. Real-World HE Libraries.*: Homomorphic encryption libraries (HElib, SEAL) use small or sparse secrets. Mitigations, such as randomized secret distributions or enforced minimum Hamming weights, should be analyzed to assess real-world impact.

d) *4. Parameter Tuning and Local Search.*: Optimizing local flips and BFS expansions in Chill is an open problem. Formalizing cost functions and iteration bounds could further reduce complexity.

e) *5. Incremental Chill Expansion.*: Dynamically adjusting Chill vs. Cool classification based on success rates may improve efficiency. Experimental validation is needed to optimize runtime and accuracy.

Implementation will build on the original Cool and Cruel framework [6].

C. Conclusion

We introduced a Cruel, Chill, Cool partition to refine LWE key recovery, balancing brute force, ML-based ranking, and bit-flipping. While Chill adds complexity, it reduces the need for excessive brute-force searches.

Future work includes tuning lattice reduction depth (ω), improving ML classification for Chill columns, and refining local search. Empirical tests up to $n = 1024+$ will evaluate feasibility against real-world encryption libraries. Combining cryptanalytic strategies with ML insights may enhance attacks on small or sparse LWE secrets.

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